

CAMBRIDGE INTERNATIONAL EXAMINATIONS

Cambridge International Advanced Subsidiary and Advanced Level

MARK SCHEME for the October/November 2015 series

9709 MATHEMATICS

9709/62

Paper 6, maximum raw mark 50

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge will not enter into discussions about these mark schemes.

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Mark Scheme Notes

Marks are of the following three types:

M Method mark, awarded for a valid method applied to the problem. Method marks are not lost for numerical errors, algebraic slips or errors in units. However, it is not usually sufficient for a candidate just to indicate an intention of using some method or just to quote a formula; the formula or idea must be applied to the specific problem in hand, e.g. by substituting the relevant quantities into the formula. Correct application of a formula without the formula being quoted obviously earns the M mark and in some cases an M mark can be implied from a correct answer.

A Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. Accuracy marks cannot be given unless the associated method mark is earned (or implied).

B Mark for a correct result or statement independent of method marks.

- When a part of a question has two or more “method” steps, the M marks are generally independent unless the scheme specifically says otherwise; and similarly when there are several B marks allocated. The notation DM or DB (or dep*) is used to indicate that a particular M or B mark is dependent on an earlier M or B (asterisked) mark in the scheme. When two or more steps are run together by the candidate, the earlier marks are implied and full credit is given.
- The symbol ∇ implies that the A or B mark indicated is allowed for work correctly following on from previously incorrect results. Otherwise, A or B marks are given for correct work only. A and B marks are not given for fortuitously “correct” answers or results obtained from incorrect working.
- Note: B2 or A2 means that the candidate can earn 2 or 0.
B2/1/0 means that the candidate can earn anything from 0 to 2.

The marks indicated in the scheme may not be subdivided. If there is genuine doubt whether a candidate has earned a mark, allow the candidate the benefit of the doubt. Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored.

- Wrong or missing units in an answer should not lead to the loss of a mark unless the scheme specifically indicates otherwise.
- For a numerical answer, allow the A or B mark if a value is obtained which is correct to 3 s.f., or which would be correct to 3 s.f. if rounded (1 d.p. in the case of an angle). As stated above, an A or B mark is not given if a correct numerical answer arises fortuitously from incorrect working. For Mechanics questions, allow A or B marks for correct answers which arise from taking g equal to 9.8 or 9.81 instead of 10.

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The following abbreviations may be used in a mark scheme or used on the scripts:

AEF	Any Equivalent Form (of answer is equally acceptable)
AG	Answer Given on the question paper (so extra checking is needed to ensure that the detailed working leading to the result is valid)
BOD	Benefit of Doubt (allowed when the validity of a solution may not be absolutely clear)
CAO	Correct Answer Only (emphasising that no “follow through” from a previous error is allowed)
CWO	Correct Working Only – often written by a ‘fortuitous’ answer
ISW	Ignore Subsequent Working
MR	Misread
PA	Premature Approximation (resulting in basically correct work that is insufficiently accurate)
SOS	See Other Solution (the candidate makes a better attempt at the same question)
SR	Special Ruling (detailing the mark to be given for a specific wrong solution, or a case where some standard marking practice is to be varied in the light of a particular circumstance)

Penalties

MR –1	A penalty of MR –1 is deducted from A or B marks when the data of a question or part question are genuinely misread and the object and difficulty of the question remain unaltered. In this case all A and B marks then become “follow through” marks. MR is not applied when the candidate misreads his own figures – this is regarded as an error in accuracy. An MR –2 penalty may be applied in particular cases if agreed at the coordination meeting.
PA –1	This is deducted from A or B marks in the case of premature approximation. The PA –1 penalty is usually discussed at the meeting.

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1	$\Sigma x - 100n = 216$ $2416 - 100n = 216$ $n = 22$ OR $\frac{2416}{n} = \frac{216}{n} + 100$ $n = 22$	B1 B1 B1 3 B1 B1 B1	$\Sigma x - 100n$ seen Subst 2416 for their Σx Correct answer 2416/n seen or 216/n + 100 oe eg $\Sigma x/n - 100 = 216/n$ correct equation Correct answer
2	P(no men) $\frac{{}^9C_6}{{}^{16}C_6} = \frac{84}{8008} = \frac{21}{2002} = \frac{3}{286}$ = 0.0105 OR $\frac{9}{16} \times \frac{8}{15} \times \frac{7}{14} \times \frac{6}{13} \times \frac{5}{12} \times \frac{4}{11} = 0.0105$	B1 B1 B1 3 B1 B1 B1	9C_6 seen anywhere ${}^{16}C_6$ seen as denom of fraction oe Correct final answer (9 × 8 × 7 × 6 × 5 × 4) seen anywhere Correct unsimplified denom Correct final answer
3 (i)	$\frac{1}{4}$	B1 1	
(ii)	$\left(\frac{3}{4}\right)^4 \left(\frac{1}{4}\right) = \frac{81}{1024} = 0.0791$	M1 A1 2	Expression of form $p^4(1-p)$ only, $p = 1/4$ or $3/4$ Correct answer
(iii)	P(all diff) = $\frac{1}{4} \times \frac{1}{4} \times \frac{1}{4} \times \frac{1}{4} \times 4!$ = $\frac{3}{32}$ (0.0938) OR $1 \times \frac{3}{4} \times \frac{2}{4} \times \frac{1}{4} = \frac{3}{32}$	M1 M1 A1 3	4! on numerator seen mult by $k \geq 1$ or $3 \times 2 \times 1$ on num oe, must be in a fraction. 4^4 on denom or 4^3 on denom with the $3 \times 2 \times 1$ Correct answer
4 (i)	Two in same taxi: ${}^6C_2 \times {}^4C_4 \times 2$ or ${}^6C_2 + {}^6C_4$ = 30	M1 M1 A1 3	6C_4 or 6C_2 oe seen anywhere 'something' ×2 only or adding 2 equal terms Correct final answer
(ii)	MJS in taxi $({}^5C_1 \times 2 \times 2) \times {}^4P_4$ = 480	M1 M1 M1 A1 4	5P_1 , 5C_1 or 5 seen anywhere Mult by 2 or 4 oe Mult by 4P_4 oe eg 4! or $4 \times {}^3P_3$ or can be part of 5! Correct final answer

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5	(i)	<table><tr><th>team A</th><th></th><th>team B</th></tr><tr><td></td><td>7</td><td>5 7 9</td></tr><tr><td>4 4 2</td><td>8</td><td>2 3 4 6</td></tr><tr><td>9 8 7 6 1</td><td>9</td><td>4 5 6</td></tr><tr><td>9 7 4 0</td><td>10</td><td>1 8</td></tr><tr><td>6 5</td><td>11</td><td>1 3 5</td></tr><tr><td>2</td><td>12</td><td></td></tr></table>	team A		team B		7	5 7 9	4 4 2	8	2 3 4 6	9 8 7 6 1	9	4 5 6	9 7 4 0	10	1 8	6 5	11	1 3 5	2	12		B1	Correct stem can be upside down, ignore extra values, allow 70, 80 etc with suitable numerical key										
		team A		team B																															
			7	5 7 9																															
		4 4 2	8	2 3 4 6																															
9 8 7 6 1	9	4 5 6																																	
9 7 4 0	10	1 8																																	
6 5	11	1 3 5																																	
2	12																																		
B1	Correct team A must be on LHS, alignment ± half a space, no late entries squeezed in, no crossing out if shape is changed																																		
B1	Correct team B in single diagram can be either LHS or RHS																																		
B1 4	Correct key or keys for their diagram/s, need both teams, at least one kg.																																		
	(ii)	LQ = 91 UQ = 109 IQ range = 18	B1 B1✓ 2	Both quartiles correct Correct IQR ft wrong quartiles, LQ < UQ, not 12 – 4 etc																															
	(iii)	$\Sigma x_{15} = 1399$ $\Sigma x_{16} = 16 \times 93.9 = 1502.4$ New wt = $1502.4 - 1399 = 103$ (103.4)	M1 M1 A1 3	Attempt at Σx_{15} for either team Mult 93.9 by 16 attempt Correct answer																															
6	(i)	<table><tr><td></td><td colspan="4">Spinner A</td></tr><tr><td></td><td>1</td><td>2</td><td>3</td><td>3</td></tr><tr><td rowspan="4">Spinner B</td><td>–3</td><td>(–2)</td><td>–1</td><td>0</td><td>0</td></tr><tr><td>–2</td><td>–1</td><td>0</td><td>(1)</td><td>1</td></tr><tr><td>–1</td><td>0</td><td>1</td><td>2</td><td>2</td></tr><tr><td>1</td><td>2</td><td>3</td><td>4</td><td>4</td></tr></table>		Spinner A					1	2	3	3	Spinner B	–3	(–2)	–1	0	0	–2	–1	0	(1)	1	–1	0	1	2	2	1	2	3	4	4	B1 1	
			Spinner A																																
			1	2	3	3																													
Spinner B	–3	(–2)	–1	0	0																														
	–2	–1	0	(1)	1																														
	–1	0	1	2	2																														
	1	2	3	4	4																														
(ii)	<table><tr><td>x</td><td>–2</td><td>–1</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>prob</td><td>$\frac{1}{16}$</td><td>$\frac{2}{16}$</td><td>$\frac{4}{16}$</td><td>$\frac{3}{16}$</td><td>$\frac{3}{16}$</td><td>$\frac{1}{16}$</td><td>$\frac{2}{16}$</td></tr></table>	x	–2	–1	0	1	2	3	4	prob	$\frac{1}{16}$	$\frac{2}{16}$	$\frac{4}{16}$	$\frac{3}{16}$	$\frac{3}{16}$	$\frac{1}{16}$	$\frac{2}{16}$	M1 M1 A1 3	Their values in (i) as the top line, seen listed in (ii) or used in part (iii) Attempt at probs seen evaluated, need at least 4 correct from their table Correct table seen																
x	–2	–1	0	1	2	3	4																												
prob	$\frac{1}{16}$	$\frac{2}{16}$	$\frac{4}{16}$	$\frac{3}{16}$	$\frac{3}{16}$	$\frac{1}{16}$	$\frac{2}{16}$																												
(iii)	$E(X) = 1$ $\text{Var}(X) = ((-2)^2 + 2 + 3 + 12 + 9 + 32)/16 - 1^2$ $= \frac{62}{16} - 1$ $= \left(\frac{23}{8}\right) (2.875)$ OR using $\Sigma p(x - \bar{x})^2 = (9 + 8 + 4 + 0 + 3 + 4 + 18)/16$ $= \frac{46}{16} = 2.875$	M1 M1 A1 3 M1 M1 A1	Attempt at $E(X)$ from their table if $\Sigma p = 1$ Evaluating $\Sigma x^2 p - [\text{their } E(X)]^2$ allow $\Sigma p \neq 1$ but all p 's < 1 Correct answer																																

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(iv)	P(even given +ve) $= \frac{5}{9}$	M1 A1 2	Counting their even numbers and dividing by their positive numbers Correct answer
	OR P(even given +ve) = $\frac{\left(\frac{5}{16}\right)}{\left(\frac{9}{16}\right)}$	M1	Using cond prob formula not P(E) × P(+ve) need fraction over fraction accept any of $\frac{5/16 \text{ or } 6/16 \text{ or } 9/16}{9/16 \text{ or } 10/16 \text{ or } 13/16}$
	$= \frac{5}{9}(0.556)$	A1	Correct answer
7 (a) (i)	$P(x > 3900) = P\left(z > \frac{3900 - 4520}{560}\right)$ $= P(z > -1.107) = \Phi(1.107)$ $= 0.8657$ Number of days = 365×0.8657 $= 315 \text{ or } 316 (315.98)$	M1 M1 A1 B1 4	Standardising no cc no sq rt no sq Correct area Φ ie > 0.5 Prob rounding to 0.866 Correct answer ft their wrong prob if previous A0, $p < 1$, ft must be accurate to 3sf
	(ii) $z = 1.165$ $1.165 = \frac{8000 - m}{560}$ $m = 7350 (7347.6)$	B1 M1 A1 3	± 1.165 seen Standardising eqn allow sq, sq rt, cc, must have z-value eg not 0.122, 0.878, 0.549, 0.810. Correct answer rounding to 7350
	(iii) $P(0, 1) = (0.878)^6 + {}^6C_1(0.122)^1(0.878)^5$ $= 0.840$ accept 0.84 Normal approx. to Binomial. M0, M0, A0	M1 M1 A1 3	Binomial term ${}^6C_x p^x(1-p)^{6-x}$ $0 < p < 1$ seen Correct unsimplified expression Correct answer
	(b) $P(< 2\mu) = P\left(z > \frac{2\mu - \mu}{\sigma}\right) = P(z < 1.5)$ $= 0.933$	M1 M1 A1 3	Standardising with μ and σ Attempt at one variable and cancel Correct answer